

Complex Systems in Bioinformatics SoSe 2026

Exercise 10

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Deadline: 23 June 2026, before the lecture

The homework should be worked out individually, or in groups of 2 students. Pen & paper exercises should be handed at the designated deadline. Each solution sheet must contain the names and Matrikel No. of all group members and the name of the group. Please staple all sheets. Programming exercises must be submitted via Whiteboard.

1. Exercise [10 points: 1+2+1+3+3 points]

Consider the system of differential equations

$$\dot{x} = A + x^2y - Bx - x$$

$$\dot{y} = Bx - x^2y$$

with the parameters $A, B > 0$ and the variables $x, y \in \mathbb{R}_{\geq 0}$.

- Determine the steady state of the system.
- Derive the Jacobi matrix of the system and give it at steady state.
- Determine the trace and determinant of the Jacobi matrix at steady state.
- Give example parameter combinations (A, B) that allow for limit cycle oscillations or a stable focus.
- Implement the model in a program of your choice. Plot time courses and a phase plane for each of the two dynamics and at least two initial conditions each. (Include your chosen parameters and initial conditions in your solution.)